

Richard Djarbeng

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PERSONAL STATEMENT

Engineer and researcher with hands-on experience developing, training, and deploying ML models across computer vision, predictive analytics, and environmental sensing domains. Proficient in Python, PyTorch, Keras, and FastAI, with practical experience deploying models on Google Cloud (Vertex AI, BigQuery ML) and Hugging Face. Strong foundation in data analysis, sensor calibration, and IoT systems, with a track record of applying ML techniques to real-world problems in air quality, image segmentation, and geospatial monitoring. Holds an MSc in Electrical and Computer Engineering from Carnegie Mellon University, with a focus on deep learning, computer vision, and IoT. Has practical experience with Google Cloud ML tooling including BigQuery ML, Vertex AI, and AutoML, applied through hands-on projects in predictive modelling, model deployment, and data pipeline development.

EDUCATION

Carnegie Mellon University

July 2022 – May 2024

MSc Electrical and Computer Engineering

Pittsburgh, PA (and Kigali, Rwanda)

- **Relevant Coursework:** Introduction to Deep Learning, Computer Vision, Internet of Things, Autonomous Driving, Data Analytics, Data Structures and Algorithms, Introduction to Information Security, Cognitive Robotics, Wireless Networks

University of Ghana

November 2021

BSc Computer Engineering

Accra, Ghana

- **Electives:** Software Engineering, Computer Graphics
- **Relevant Coursework:** Object Oriented Programming, C/C++ Programming, Computational Tools, Robotics and Computer Vision, Data Structures and Algorithms, Introduction to Artificial Intelligence, Computer Architecture, Operating Systems

CERTIFICATIONS

AWS Certified Solutions Architect – Associate

June 2026

Amazon Web Services

- **Designing Secure Architectures:** Demonstrated ability to build secure AWS infrastructure using identity management (IAM), data encryption, and network security controls to protect applications and sensitive data.
- **Designing Resilient Architectures:** Proven skills in creating highly available and fault-tolerant systems using Multi-AZ deployments, Auto Scaling, load balancing, and disaster recovery strategies.
- **Designing High-Performing Architectures:** Skilled at selecting appropriate compute, storage, and networking resources, including caching solutions, to deliver fast and scalable applications.
- **Designing Cost-Optimized Architectures:** Expertise in optimizing costs through right-sizing resources, choosing appropriate storage classes, and leveraging flexible pricing models such as Spot Instances.

PROFESSIONAL DEVELOPMENT

Google Get Certified Program

June 2024 – August 2024

Professional Machine Learning Engineer (PMLE) Track

Google Cloud

- Completed intensive instructor-led training and hands-on "arcade" labs with a Google instructor.
- Collaborated with international cohorts across multiple time zones to complete practical machine learning engineering exercises.
- Gained applied experience in production ML workflows, including model training, deployment, monitoring, and MLOps using Vertex AI, AutoML, and BigQuery ML.

EXPERIENCE

Carnegie Mellon University

July 2024 – October 2025

Research Associate

Kigali, Rwanda

- **Environmental Sensor Development:** Designed and developed low-cost sensor nodes for measuring temperature, humidity, wind speed, and environmental parameters using innovative AI-driven calibration approaches.
- **Sensor Testing and Calibration:** Applied machine learning techniques to improve the accuracy and reliability of low-cost sensors, benchmarking against high-grade reference instruments.
- **Data Storage and Management:** Collaborated with principal investigators and external researchers to design a secure data storage and management framework, ensuring collected environmental data was accessible to project stakeholders.
- **System Design:** Designed and built data management systems and web-accessible dashboards for environmental sensor networks across the full development lifecycle.
- **Academic Paper Preparation:** Collaborated with the research team to prepare and publish academic papers documenting the project's findings and methodologies.

Carnegie Mellon University

January 2024 – May 2024

Research Assistant

Kigali, Rwanda

- **Interdisciplinary Research:** Worked with the mechanical engineering department to research alternative methods of predicting biomass pollutants in the air using sensor data from aethalometers.
- **Air Quality Prediction:** Improved existing low-cost means of predicting black carbon pollutants using RGB pictures of filters to distinguish pollution from wood burning versus fossil fuel combustion.
- **Pollution Source Classification:** Developed computer vision models to classify pollution sources from particulate matter filters using smartphone imagery, enabling low-cost air quality source apportionment. Research featured on the [CMU Africa](#) website.
- **Data Analysis and Experimentation:** Conducted data analysis on datasets collected over more than 2 years and verified experiments to measure black carbon pollutants.

Carnegie Mellon University

August 2023 – December 2023

Teaching Assistant, Internet of Things

Kigali, Rwanda

- **Teaching and Mentoring:** Conducted recitations for the Internet of Things (IoT) course, focused on building IoT systems with web APIs using JavaScript and Node.js on the Raspberry Pi.
- **Academic Support:** Assisted students in understanding circuit design, embedded systems programming, REST, HTTP, and wireless connectivity (2G, 3G).

AfriqAir - Kigali Collaborative Research Center

May 2023 – August 2023

Graduate Intern

Kigali, Rwanda

- **Web Dashboard Development:** Developed a production-grade web dashboard using React.js and TypeScript for real-time visualisation and interpretation of air quality data, consuming live sensor feeds.
- **Prototype Development:** Built a prototype IoT system to automate the diagnostic and data collection process from the Thermofisher TEOM air quality system using Python and C++ for Raspberry Pi and Arduino.
- **System Design:** Designed system specifications and architecture for an IoT wireless data logging unit, coordinating requirements across hardware and software teams.

Sorphise GH Ltd

July 2020 – January 2021

Intern

Ghana

- **Web Application Development:** Developed and deployed web applications using JavaScript, PHP, React (GatsbyJS), Netlify, and NetlifyCMS.
- **CMS Integration and Architecture:** Led the adoption of a headless CMS architecture decoupling content from code using NetlifyCMS, integrated with GitHub for CI/CD.
- **Site Deployment and Management:** Deployed sites on Netlify, configured CMS integrations, resolved DNS errors, and monitored performance on live websites.

Promasidor GH Ltd

June 2019 – August 2019

Intern

Accra, Ghana

- Provided hardware and software support to employees and assisted with network troubleshooting on switches and routers.

Inlaks GH Ltd

June 2018 – August 2018

Intern

Accra, Ghana

- Assisted engineers with maintenance and installation of applications on ATMs; tested hardware modules and troubleshooted faulty sensors and keypads.

TECHNICAL SKILLS

Machine Learning and AI: PyTorch, Keras, FastAI, Scikit-learn, XGBoost, Logistic Regression, ResNet, CNN architectures, Transfer Learning, Model Deployment

Computer Vision: Object Detection (YOLO-World), Image Segmentation (U-Net, ISNet, Silueta, U2Net), Deepfake Detection, Semantic Segmentation, Satellite Image Processing, Roboflow

Cloud and MLOps: Google Cloud Platform (Vertex AI, AutoML, BigQuery ML, Dataflow, Looker Studio), AWS, Hugging Face Spaces, CI/CD, Model Endpoint Deployment

Data and Analytics: Python (Pandas, NumPy), SQL, BigQuery, Data Visualisation, Air Quality Data Analysis, Geospatial Analysis (Google Earth Engine), Feature Engineering

Embedded and IoT: C, C++, Raspberry Pi, Arduino, GSM Modules, Sensor Calibration, Wireless Connectivity (2G, 3G)

Web and Deployment: React.js, Node.js, TypeScript, JavaScript (ES6+), RESTful APIs, Gradio, Netlify, Render

Languages: Python, JavaScript, Java, C++, C#, SQL, PHP, Ruby, Dart (Flutter)

Tools and Practices: Git, Jupyter Notebooks, Test-Driven Development, Agile methodologies

LEADERSHIP EXPERIENCE

IoT Club - Carnegie Mellon University Africa

2022 – 2024

Club President

Kigali, Rwanda

- Led a team of 6 committee members to organise campus events enhancing student knowledge of IoT, embedded systems, and AI applications.
- Orchestrated field trips to Zipline, a company using autonomous drones for medical supply delivery in Rwanda, providing students with real-world exposure to applied AI and robotics in humanitarian contexts.
- Served as a technical advisor for student projects in data collection, GPS integration, and microcontroller programming.

PROJECTS

Background Remover

August 2024

Machine Learning Engineer

Hugging Face Space

- Deployed a semantic segmentation model for HD background removal using the **rembg** library and a production-ready Gradio interface on Hugging Face Spaces.
- Supports multiple model architectures (U-Net, ISNet, Silueta, U2Net) to cover different segmentation workflows, with options for solid-colour or transparent backgrounds, alpha matting, post-processing mask, and mask-only output.
- Exports PNG or WEBP in full HD at no cost, unlike most tools that charge for HD downloads.
- *Technologies: Python, Deep Learning, Computer Vision, Gradio, rembg, CI/CD*

Air Quality Analysis Dashboard

May 2024

ML Researcher and Frontend Developer

airanalysis.netlify.app

- Developed an interactive dashboard to analyse $PM_{2.5}$ trends in Kigali, leveraging US Embassy reference data (AirNow.gov) and planetary boundary layer dynamics to identify diurnal pollution patterns.
- Quantified public health risks by identifying that 57% of readings exceeded WHO "Unhealthy" thresholds, with an average concentration of $43.31 \mu g/m^3$.
- Engineered multi-city comparative analysis modules and visual context features (visibility degradation) to communicate environmental impacts to stakeholders.
- Collaborated with academic advisors to validate model-observation evaluations and emission source meteorology.
- *Technologies: React.js, TypeScript, Data Visualisation, Air Quality Modelling*

Predicting Loan Risk with AutoML on Google Cloud Vertex AI

May 2024

Machine Learning Engineer

Google Cloud

- Developed a machine learning model to predict loan default risk using Google Cloud Vertex AI AutoML.
- Engineered features from tabular data, optimised model training within allocated compute budgets with early stopping to prevent overfitting.
- Evaluated model using precision/recall curves, ROC curves, and Explainable AI for feature importance analysis.

- Deployed model endpoints and integrated with the Shared ML Service, demonstrating end-to-end ML lifecycle management.
- *Technologies: Google Cloud Vertex AI, AutoML, BigQuery, Python, Explainable AI*

E-commerce Purchase Prediction with BigQuery ML

May 2024

ML Engineer and Data Analyst

Google Cloud

- Built a high-performance classification model (XGBoost and Logistic Regression) within BigQuery ML to predict returning customer purchase intent from Google Analytics datasets.
- Improved model performance from 0.72 to **0.94 ROC AUC** through rigorous feature engineering of checkout progress, traffic sources, and device-specific behavioural data.
- Delivered actionable business intelligence: targeting the top 6% of predicted visitors could yield a **9x increase in marketing ROI**, capturing 50% of all potential conversions.
- Executed hyperparameter tuning and model selection, applying Google Cloud ML best practices for classification model optimisation.
- *Technologies: BigQuery ML, XGBoost, SQL, Feature Engineering, Marketing Analytics*

Detection and Identification of Manipulated Facial Images

September 2023 – December 2023

Deep Learning Engineer

- Developed a robust system using deep learning to detect and identify manipulated facial images, targeting deepfakes, Face Swap, and Face2Face manipulations.
- Utilised a Block-based CNN and ResNet-34 to achieve high accuracy in classifying manipulated images extracted from videos.
- Fine-tuned ResNet-50 as an additional classification backbone, applying transfer learning on facial manipulation datasets.
- *Technologies: Python, Deep Learning, Computer Vision, ResNet, CNN, PyTorch*

Pollution Source Classification from PM Filters

2024

Computer Vision Researcher

- Developed computer vision models to classify pollution sources (wood burning vs. fossil fuels) from RGB images of particulate matter filters captured via smartphone.
- Enabled low-cost source apportionment of black carbon pollutants, reducing reliance on expensive laboratory analysis.
- Research published and featured on the [CMU Africa website](#).
- *Technologies: Python, Computer Vision, Roboflow, Machine Learning, PyTorch*

YOLO-World Promptable Video Object Detection

2024

Computer Vision Engineer

- Explored YOLO-World models for promptable object detection and segmentation on real-world video footage recorded in Kigali, Rwanda.
- Built a notebook-based demo to visualise detection and segmentation results, evaluating the model family for practical deployment use cases.
- *Technologies: Python, YOLO-World, Computer Vision, Object Detection, Image Segmentation*

Transmission Corridor Monitoring using Satellite Imagery

September 2020 – August 2021

Remote Sensing Engineer

- Developed a web application system for GRIDCo/ECG in Ghana to monitor power transmission corridors using satellite imagery, deployed as a Google Cloud application.
- Applied image processing and geospatial analysis using Google Earth Engine and CNNs to detect and alert on potential hazards or encroachments.
- Demonstrated ability to work with large-scale geospatial data and domain-specific remote sensing challenges.
- *Technologies: Python, Google Earth Engine, CNN, Satellite Image Processing, Geospatial Information Systems*

Real-Time Taxi Fleet Management

2023

Data Engineer

Google Cloud

- Designed a streaming data pipeline using Google Cloud Dataflow and BigQuery to ingest and analyse real-time taxi data.
- Displayed live operational metrics on Looker Studio for real-time fleet monitoring.
- *Technologies: Google Cloud Dataflow, BigQuery, Looker Studio, Streaming Data*

IoT Security System

November 2022 – December 2022

IoT Engineer

- Engineered an IoT security system using Raspberry Pi, sensors, and Node.js for real-time monitoring.
- Implemented a WebSocket-based server for instant alerts and data visualisation.
- *Technologies: Node.js, Raspberry Pi, WebSockets, Real-time Systems*

GSM HTTP Requests

2023

Embedded Systems Engineer

[GitHub](#)

- Implemented HTTP GET and POST requests using GSM modules in C++, a core component for remote IoT data transmission from field-deployed sensor nodes.
- *Technologies: C++, Embedded Systems, GSM, IoT*

Personal Website and Blog

Ongoing

Owner and Developer

[rdjarbeng.com](#)

- Personal technical blog built with Ruby-Jekyll and SveltiaCMS, used to document ML experiments, research findings, and technical writing. Crossed 10,000 pageviews.
- *Technologies: Ruby, Jekyll, SveltiaCMS, Git, CI/CD | [GitHub](#)*

Countdown Progressive Web Application (PWA)

December 2021

Project Owner and Lead Developer

[rcountdown.netlify.app](#)

- Feature-rich PWA with themes, dark/light mode, and offline support via service workers. Attracted 90+ pull requests through open-source collaboration.
- *Technologies: JavaScript, HTML, CSS, PWAs, Git, CI/CD | [GitHub](#)*

University Application Management System

April 2020 – June 2020

Java Developer

[GitHub](#)

- Desktop application for university admissions with user registration, course and residence hall applications, status tracking, and an administrator control panel. Built with Java, JavaFX, JDBC, and SQL.
- *Technologies: SQL, Java, JavaFX, JDBC*

2D Game: Escape the Covid Maze

2020

Game Developer

- Unity3D 2D maze game where villain agents pursue the player using the A* search algorithm for pathfinding, across two levels of increasing difficulty.
- *Technologies: C#, Unity3D, A* Pathfinding*

Machine Learning Complete

Ongoing

Open Source Contributor

[GitHub](#)

- Contributor to the Machine Learning Complete educational package, a comprehensive open-source resource for democratising AI education.
- *Technologies: Python, Machine Learning, Open Source*